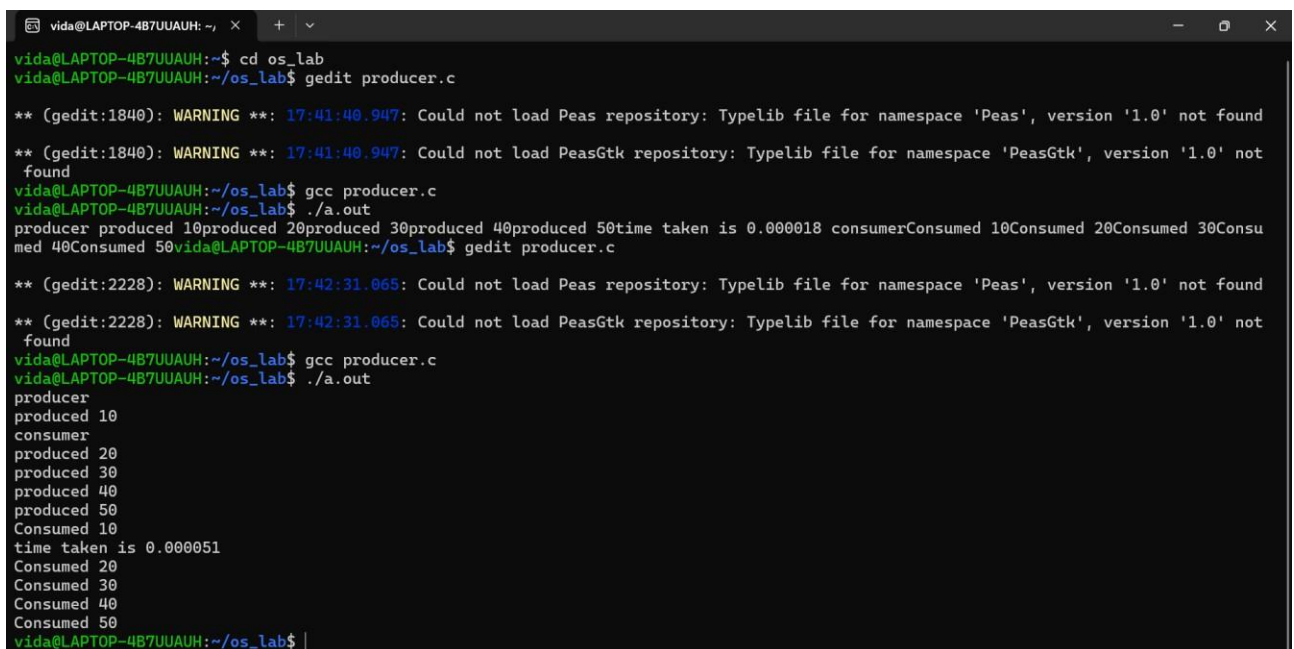


Practical-5

- A producer-consumer program was created using an anonymous pipe.
- The parent process acted as the producer and generated the values 10, 20, 30, 40 and 50.
- The child process acted as the consumer and successfully received the same values through the pipe.
- The program measured the communication time using `clock()`.
- The output showed that the data produced by the parent was successfully consumed by the child process.
- The communication time displayed in the executed output was 0.000051 seconds.

Screenshot of the Output



```
vida@LAPTOP-4B7UUAUH: ~$ cd os_lab
vida@LAPTOP-4B7UUAUH: ~/os_lab$ gedit producer.c

** (gedit:1840): WARNING **: 17:41:40.947: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:1840): WARNING **: 17:41:40.947: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
vida@LAPTOP-4B7UUAUH: ~/os_lab$ gcc producer.c
vida@LAPTOP-4B7UUAUH: ~/os_lab$ ./a.out
producer produced 10produced 20produced 30produced 40produced 50time taken is 0.000018 consumerConsumed 10Consumed 20Consumed 30Consumed 40Consumed 50
vida@LAPTOP-4B7UUAUH: ~/os_lab$ gedit producer.c

** (gedit:2228): WARNING **: 17:42:31.065: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2228): WARNING **: 17:42:31.065: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
vida@LAPTOP-4B7UUAUH: ~/os_lab$ gcc producer.c
vida@LAPTOP-4B7UUAUH: ~/os_lab$ ./a.out
producer
produced 10
consumer
produced 20
produced 30
produced 40
produced 50
Consumed 10
time taken is 0.000051
Consumed 20
Consumed 30
Consumed 40
Consumed 50
vida@LAPTOP-4B7UUAUH: ~/os_lab$ |
```

Second Task: Shell Command Execution through exec

- Implemented a C program to execute the equivalent of the shell command: `ls -l | grep ".c"`
- The program creates a pipe and uses `fork()` to create a child process.
- The `dup2()` redirects the required standard input/output streams, and `exec()` runs the commands so that the output of `ls -l` is passed to `grep ".c"`.
- The attached terminal screenshot shows the resulting list of C source files.

Screenshot of Execution Output

```
vida@LAPTOP-4B7UUAUH:~/os_lab$ gcc pipeextask.c
vida@LAPTOP-4B7UUAUH:~/os_lab$ ./a.out
-rw-r--r-- 1 vida vida 324 Sep  1 05:22 SMprocess1.c
-rw-r--r-- 1 vida vida 242 Sep  1 05:18 SMprocess2.c
prw-r--r-- 1 vida vida  0 Sep 10 06:07 client-server
-rw-r--r-- 1 vida vida  84 Aug 11 05:11 exec3.c
-rw-r--r-- 1 vida vida  72 Aug 11 04:53 execl1.c
-rw-r--r-- 1 vida vida 102 Aug 11 04:59 execlp2.c
-rw-r--r-- 1 vida vida 291 Aug 11 05:47 execcpipe.c
-rw-r--r-- 1 vida vida 242 Aug 11 05:41 exectask.c
-rw-r--r-- 1 vida vida  85 Aug 11 05:25 execv.c
-rw-r--r-- 1 vida vida 280 Sep 15 06:00 malloc.c
-rwxr-xr-x 1 vida vida 26584 Sep 22 05:17 malloc_asan
-rw----- 1 vida vida  85 Sep 15 05:57 mallocfloat.c.save
-rw-r--r-- 1 vida vida 217 Aug  8 09:32 pipe6.c
-rw-r--r-- 1 vida vida 329 Aug  8 09:42 pipe7.c
-rw-r--r-- 1 vida vida 541 Aug  8 10:19 pipe7.c.save
-rw-r--r-- 1 vida vida 246 Aug 11 06:14 pipeextask.c
-rw-r--r-- 1 vida vida 249 Aug 11 08:34 pipeextask2.c
-rw----- 1 vida vida  70 Aug  7 17:45 pipefork.c.save
-rw-r--r-- 1 vida vida 286 Aug 23 16:52 prac1.c
-rw-r--r-- 1 vida vida 388 Aug 23 18:51 prac3.c
-rw-r--r-- 1 vida vida 842 Sep 10 06:07 prac6.c
-rw-r--r-- 1 vida vida 833 Sep 23 17:43 producer.c
-rw-r--r-- 1 vida vida 312 Aug 27 05:09 segmentation.c
prw-r--r-- 1 vida vida  0 Sep 10 06:07 server-client
-rw-r--r-- 1 vida vida 239 Aug 31 08:18 signal.c
-rw-r--r-- 1 vida vida 713 Aug 31 08:40 signal5.c
-rw-r--r-- 1 vida vida 333 Sep 11 10:04 skill2.c
-rw-r--r-- 1 vida vida 286 Sep 15 04:00 sleepexit.c
-rw-r--r-- 1 vida vida 426 Sep 23 17:15 sleepexit1.c
-rw-r--r-- 1 vida vida 377 Sep 23 17:11 zombie.c
```